

	1	10	20	30	40	50	60
escherichia	MA	IDENKQKALAAA	LGQIEKQF	GKGSIMRLGE	DRSMDVET	ISTGSL	LDIALGAGGLPMG
Salmonella	MA	IDENKQKALAAA	LGQIEKQF	GKGSIMRLGE	DRSMDVET	ISTGSL	LDIALGAGGLPMG
helicobacter	MA	INEDKQKAI	SLAIKQIDKVF	GKGA	LVRLGD	KQVEKID	SI
Staphylococcus	..	MDNDRQKAL	DTVTKNMEKSF	GKGA	VMKLGDN	IGRRVST	TTSTGSLVTL

	70	80	90	100	110	120
escherichia	RIVEIYGP	ESSGKTTLT	LQVIAAAQ	REGKTC	AFIDAEHALDPI	YARKLGVDIDNLLCSQP
Salmonella	RIVEIYGP	ESSGKTTLT	LQVIAAAQ	REGKTC	AFIDAEHALDPI	YARKLGVDIDNLLCSQP
helicobacter	RIEIIYGP	ESSGKTTLS	LHIIAECQ	KNGGVCA	AFIDAEHALDVH	YARKLGVDIDNLLCSQP
Staphylococcus	RIEIIYGP	ESSGKTTV	ALHAI	AEVQSN	GVA	AFIDAEHALDPI

	130	140	150	160	170	180
escherichia	DTGEQALEIC	DALARS	GAVDVIV	VDSVAALTPKAEIE	GEIGDSH	MGLAARMMSOAMRKLA
Salmonella	DTGEQALEIC	DALARS	GAVDVIV	VDSVAALTPKAEIE	GEIGDSH	MGLAARMMSOAMRKLA
helicobacter	DTGEQALEILE	TTIRSG	IDLVVDSVAALTPKAEIE	GDMDQHVGL	QARLMSHALRKIT	
Staphylococcus	DHGEQGLEIA	EAFVRS	GAVDVIV	VDSVAALTPKAEIE	GEIGDM	GDTHVGLQARLMSQALRKLS

	190	200	210	220	230
escherichia	GNLKSNTLL	IFINQIRMKIC	VM.FGNPETT	TGGNALKFYASVR	LDIRRI
Salmonella	GNLKSNTLL	IFINQIRMKIC	VM.FGNPETT	TGGNALKFYASVR	LDIRRI
helicobacter	GNLKSNTLL	IFINQIRMKIC	MMGYGSPETT	TGGNALKFYASVR	LDIRRI
Staphylococcus	GAISKNTTA	IFINQIREK	VM.FGNPETT	PGRALKFYSSVR	LEVRR

	240	250	260	270	280	290
escherichia	GSETRVKV	VKNKIAA	PFKQAEFQ	ILYGEGIN	FY	GELVDLG
Salmonella	GSETRVKV	VKNKIAA	PFKQAEFQ	ILYGEGIN	FY	GELVDLG
helicobacter	GNRAKAKV	VKNKIAA	PFRFAEFQ	IMFGEGIS	KEGEI	IDYGVKLD
Staphylococcus	GNRTKIKV	VKNKIAA	PFRFAEFQ	IMYGQGIS	KEGEI	IDYGVKLD

	300	310	320	330	340	350
escherichia	GQGKANATA	WLNKDN	ETAKET	EKKVREL	LLSNPNST	PDFSVDDSE
Salmonella	GQGKANATA	WLNKDN	ETAKET	EKKVREL	LLSNPNST	PDFSVDDSE
helicobacter	GQGRENAK	AL	LKEDK	ALANE	ITL	KIKESIGS
Staphylococcus	GQGRENV	KMYLNKDN	QIKEE	IDRKLRE	KLGIS	DGDVE